



Society of Petroleum Engineers

SPE-230199-MS

Oil Field Unlocks Additional Oil Handling Capacity Through Off Plant De-Gassing Opportunities

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230199-MS](https://doi.org/10.2118/230199-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The proposed degassing solution seeks to enhance oil handling capacity by establishing an off-plant separation facility at the pad or mini-pad level. This will facilitate the flowback of shut-in strings and optimize the performance of choked back producers. The scope includes the deployment of a portable separator at the sub-station level, directing gas to a nearby depletion network, and routing oil to the gathering system.

For proof of concept, a test separator was hooked up on a pad using existing test headers in one of the brown oil fields in the Middle East. A temporary line was laid down from separator at oil pad to nearby gas producer using flange to flange piping. The operation started by diverting one well through separator and then more wells were added progressively, starting with those having the highest GOR. The setup followed HAZOP, dispersion Modelling, and pressure testing protocols. The operations procedure and data logging were coordinated across teams. Real time monitoring allowed phased execution while ensuring operational compliance. The initial phase of the operation successfully diverted 25 MMSCFD of gas through a localized pad level separation system. This intervention unlocked approximately 7000 bbl/day by enabling flowback of previously shut-in wells and optimizing chokes on active producers. All startup activities, including pressure testing, flare commissioning, and separator integration, were completed without operational disruptions. The project was executed within planned timelines and met all safety and compliance requirements.

The localized degassing approach proved effective in relieving gas handling bottlenecks without requiring modifications to central processing facilities. Real time monitoring and phased well addition strategy allowed flexible and adaptive execution. Close coordination between operations & planning teams ensured smooth implementation, with alignment maintained across stakeholders throughout the project lifecycle. Pad level degassing offers a scalable, low-cost solution to manage surface constraints in brownfield environments. The project demonstrated how temporary infrastructure can be leveraged to unlock oil potential using existing resources. This approach minimizes investment while delivering tangible production gains and can serve as a replicable model for other constrained assets facing similar limitations.

This work offers a proven method to unlock production in constrained assets using pad level degassing. It demonstrates how to repurpose existing systems for quick, scalable gains minimizing investment while maintaining safety and efficiency. The model is transferable across mature fields with similar infrastructure challenges.

Introduction

As oil reservoirs reach maturity, water cut typically increases, necessitating expanded gas lift operations. Concurrently, certain reservoirs experience higher gas-oil ratios (GOR), resulting in additional gas entering the production network and creating bottlenecks that can restrict oil output. Robust management of associated gas is essential for optimizing oil recovery from mature fields. This strategy aims to reduce reliance on downstream gas processing infrastructure, enable the efficient operation of high-GOR wells, and improve operational flexibility—particularly during periods of elevated demand for oil production.

Oil production from Brown Fields in the Middle East presents a strategic opportunity for value enhancement through targeted improvements in gas handling infrastructure. As the Gas-Oil Ratio (GOR) continues to rise, capital investments in expanding gas processing capacity and debottlenecking transfer lines via new infrastructure initiatives are expected to yield sustainable, long-term returns. In the short term, leveraging available ullage within non-associated gas (NAG) networks enables monetization of latent capacity, unlocking incremental oil production and strengthening asset-level cash flow.

To support this strategy, rental test separators have been deployed at remote degassing stations (RDS) and mini-pads, offering a tactical, asset-light solution for redirecting associated gas into NAG networks. This approach facilitates the reactivation of shut-in wells and optimizes existing production assets, thereby improving throughput and enhancing operational efficiency within current facility constraints. As part of the proof-of-concept phase, a test separator was installed on a production pad using existing test headers. A temporary flange-to-flange pipeline was laid from the separator to a nearby gas producer. Operations commenced with a single well being diverted through the separator, followed by the gradual inclusion of additional wells prioritized by GOR. The setup adhered to HAZOP, dispersion modeling, and pressure testing protocols. Operational procedures and data logging were coordinated across multidisciplinary teams, with real-time monitoring enabling phased execution while maintaining full compliance with safety and operational standards.

The initial phase successfully diverted approximately 25 MMSCFD of gas and unlocked an estimated 7,000 BOPD. All startup activities—including pressure testing, flare commissioning, and separator integration—were completed without incident. The project was delivered within the planned schedule and met all safety, environmental, and regulatory requirements. Strategic deployment of rental well test separators as localized degassing solutions has the potential to unlock over \$2 billion in incremental value. This efficient and innovative integration of associated and non-associated gas networks is expected to deliver substantial commercial impact across the asset portfolio, reinforcing the viability of scalable, field-level optimization strategies.

Concept Overview

In one of the mature brown fields located in the Middle East, the oil and gas production network is designed around a centralized processing philosophy. Fluids are produced from multiple wells distributed across pads and mini-pads. These wells tap into various reservoir formations and deliver a multiphase stream—comprising oil, gas, and water. These wells are grouped into pads and mini-pads, which collect the multiphase fluids—oil, gas, and water—and route them through flowlines to one of ten Remote Degassing Stations (RDS). Some of these RDSs perform initial separation before forwarding the stream to a Central Degassing Station (CDS), where final separation occurs. Stabilized crude oil is then pumped via the Main

Oil Line (MOL) to export terminals, while gas is routed to non-associated gas facilities for processing, and produced water is either reinjected or disposed of through Water Disposal Wells (WDW).

In a mature brown field located in the Middle East, the deployment of Multiphase Flow Meters (MPFMs) is a cornerstone of the production monitoring and well testing strategy. These flowmeters are installed at various levels of the network—minipads, pads, and Remote Degassing Stations (RDS)—to enable real-time measurement of oil, gas, and water rates without requiring phase separation. This setup supports efficient diagnostics, production allocation, and reservoir performance tracking.

Minipad-Level Configuration

At the minipad level, each cluster typically connects 7–10 oil-producing wells through a shared production header. A single MPFM is installed at the test header, allowing one well at a time to be diverted for testing. This arrangement facilitates rapid well diagnostics and supports calibration of production data. Additionally, portable separators can be hooked up at the MPFM header to validate readings and conduct two-phase separation when necessary, especially for wells with high gas-oil ratios (GOR).

Pad-Level Configuration

Pads accommodate a larger number of wells with 2 test headers containing flowmeters at each test header. For instance, a pad connecting 20 wells uses a shared header with MPFM at each test header for sequential well testing. Portable separators are also deployed at pad-level test headers to bypass the flowmeters when needed, particularly for calibration or when dealing with high-GOR wells. This flexibility ensures accurate flow measurements and supports production optimization efforts.

RDS-Level Configuration

Remote Degassing Stations are equipped with multiple MPFMs based on the number of connected wells. The standard design follows a ratio of one flowmeter per 12 producers, with each meter capable of conducting up to 25 tests per month. Some RDSs are equipped with additional meters to support expanded testing capacity. In cases where wells exhibit high water cut or elevated GOR, test separators are used alongside MPFMs to ensure precise measurement and flow assurance.

The MPFM infrastructure plays a critical role in enabling the proposed localized degassing strategy. The degassing solution involves deploying portable separators at pad or minipad levels to remove gas before the multiphase stream reaches the RDS. These separators are hooked up directly to the MPFM test headers, leveraging existing infrastructure to minimize setup time and cost. By separating gas early in the flow path and routing it to nearby depletion network compressor stations, this approach reduces backpressure, enhances flow assurance, and supports reactivation of shut-in wells. The integration of MPFMs with portable separators ensures accurate monitoring and validation of the degassed stream, reinforcing the reliability of the solution.

A critical challenge in this field is the increasing Gas-Oil Ratio (GOR), which has begun to limit oil production due to rising backpressure in the RDS-CDS transfer lines. As GOR increases, the volume of gas in the multiphase stream grows disproportionately, creating bottlenecks that restrict the flow of oil and reduce overall system efficiency. To address this, a localized degassing solution has been proposed and piloted.

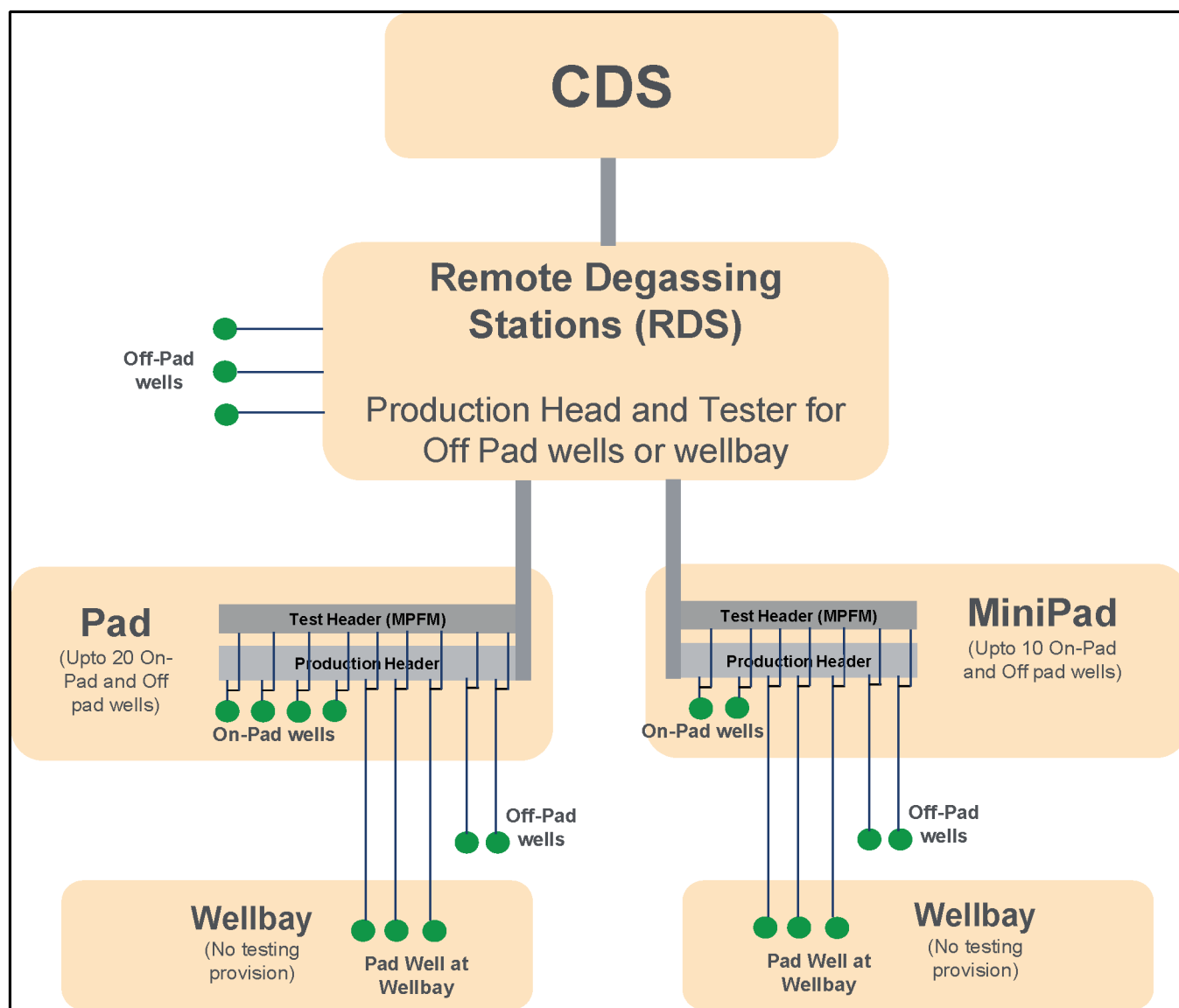


Figure 1—Production System Network at Brown Field in Middle East

The deployment of Multiphase Flow Meters across minipads, pads, and Remote Degassing Stations (RDS) not only enables efficient well testing but also serves as a critical interface for implementing localized degassing solutions. Each MPFM is installed at a test header, allowing individual wells to be diverted for real-time measurement of oil, gas, and water rates. This infrastructure is now being leveraged to support the hook-up of portable separators, which are used to remove gas from the multiphase stream before it reaches the RDS.

The separated gas from these portable separators is routed to a nearby depletion gas network, typically connected to a non-associated gas producer. This network acts as a sink for the excess gas, allowing it to be reinjected or processed without burdening the central oil handling system. By diverting gas to this depletion network, the system effectively reduces backpressure in the RDS-CDS transfer lines, which is essential in a field experiencing a rising Gas-Oil Ratio (GOR). This trend has increasingly constrained oil throughput, making early gas removal a necessary intervention.

Meanwhile, the liquid phase (oil and water) from the separator is redirected back into the conventional oil network, continuing its journey through the RDS and CDS for final separation and export. This dual-routing

strategy—gas to the depletion network and liquids to the oil network—ensures optimal flow assurance and supports the reactivation of wells previously shut-in due to high GOR.

By coupling MPFM infrastructure with portable separators and integrating them into the nearby depletion gas network, this mature brown field in the Middle East demonstrates a scalable and efficient model for managing GOR challenges. The approach not only enhances production reliability but also aligns with broader upstream innovation goals across the region.

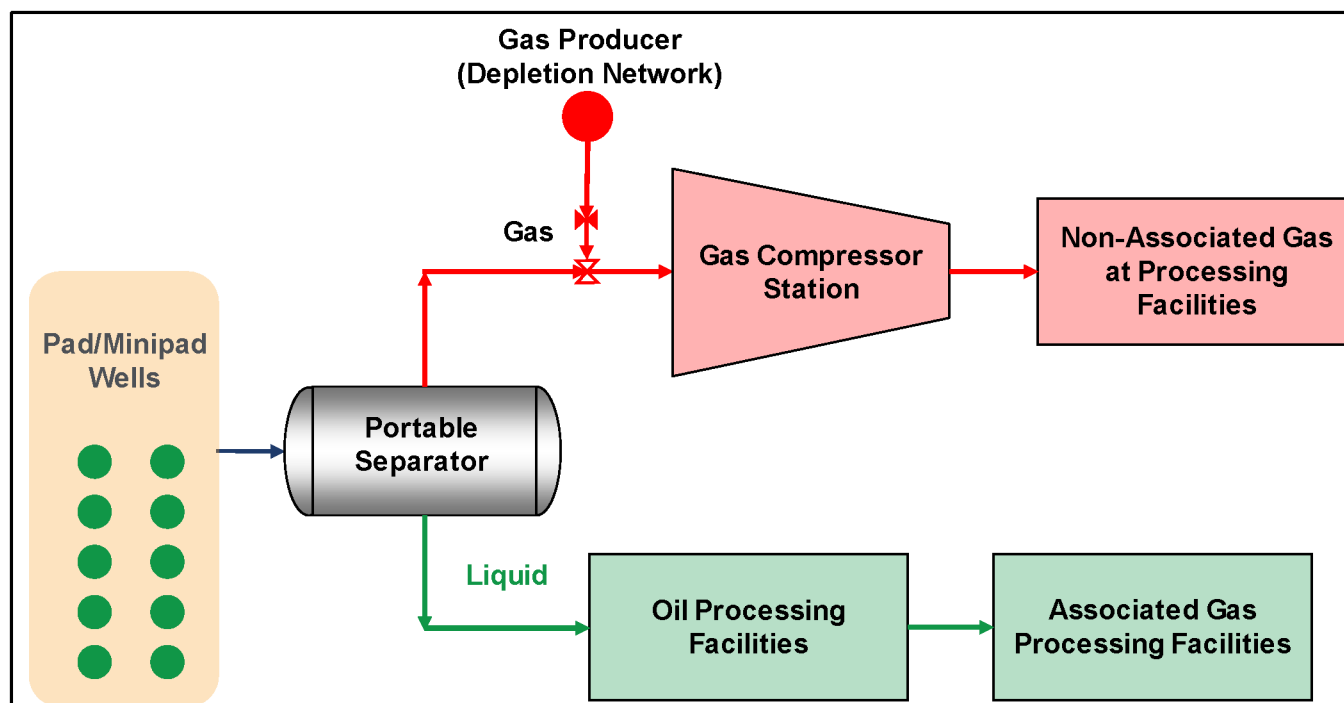


Figure 2—Overall Scheme for off Plant Degassing at Pad & Minipad

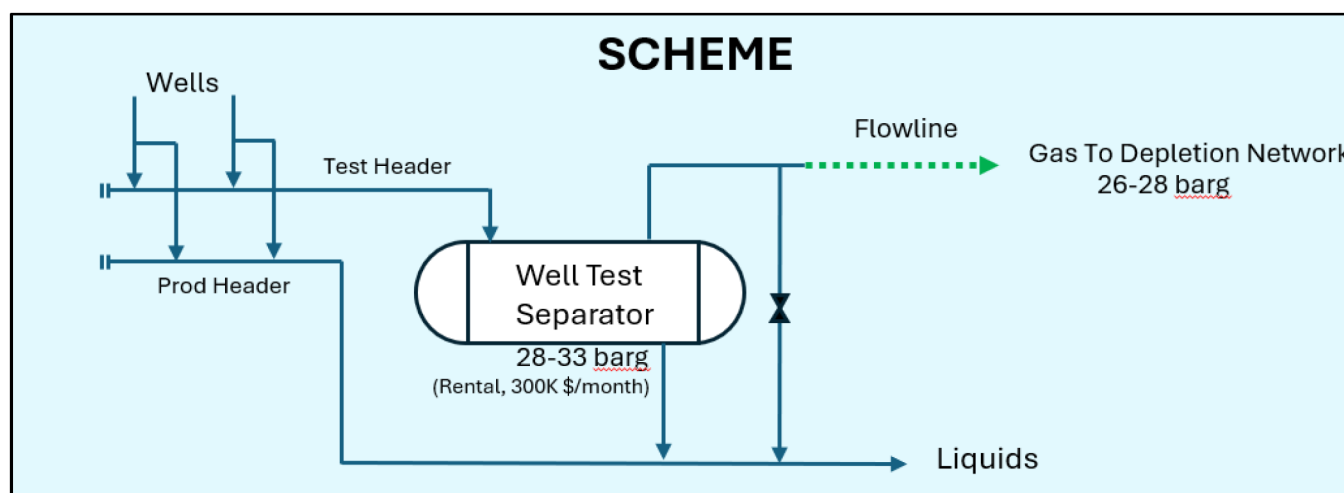


Figure 3—Proposed Method for Separator Hook at Pad & Minipad

Integrated Execution Framework

The execution methodology adopted for the proposed degassing solution followed a structured and systematic approach, ensuring technical robustness, operational safety, and strategic alignment. It began with a rigorous selection process, where Minipad-112 was chosen based on its high gas-oil ratio, production

volume, and proximity to a nearby depletion gas network. This was followed by the development of a comprehensive scope of work, detailing equipment layout, flow routing, and operational constraints. A compositional network model was then built using GAP (Petroleum Experts) to simulate various well combinations and choke settings, enabling sensitivity analysis and optimization of gas-liquid separation strategies. To ensure safety and environmental compliance, the project underwent detailed HAZID and HAZOP workshops, along with Gas Dispersion and Radiation Studies, which defined exclusion zones and emergency protocols. The on-site execution was carried out under strict Company HSE standards, with 24-hour supervision by a competent and certified crew, ensuring real-time monitoring and adherence to safety protocols. Throughout the operation, continuous data acquisition and performance evaluation enabled dynamic optimization, allowing the team to adapt the solution to varying field conditions and maximize its operational value.

Selection Criteria for Pilot Deployment

The brown field under study comprises 15 pads and minipads, each contributing varying volumes of oil and associated gas. Among these, Minipad-A was selected for the proof-of-concept deployment of the portable degassing solution based on the following strategic criteria:

- **High Gas Production:** Minipad-112 exhibited the highest Gas-Oil Ratio (GOR) among all locations, producing an average of 15,491 STB/D of oil and 58.35 MMSCFD of gas, resulting in a GOR of 3,766 scf/bbl—significantly higher than other pads.
- **Proximity to Depletion Gas Network:** The pad is located just 300 meters from the Well-X Gas Producer, which is connected to the gas depletion network. This proximity enabled a direct and efficient routing of separated gas to the non-associated gas handling infrastructure.
- **Operational Feasibility:** The pad's layout and existing infrastructure allowed for easy integration of a portable separator at the MPFM test header, facilitating two-phase separation without major modifications.

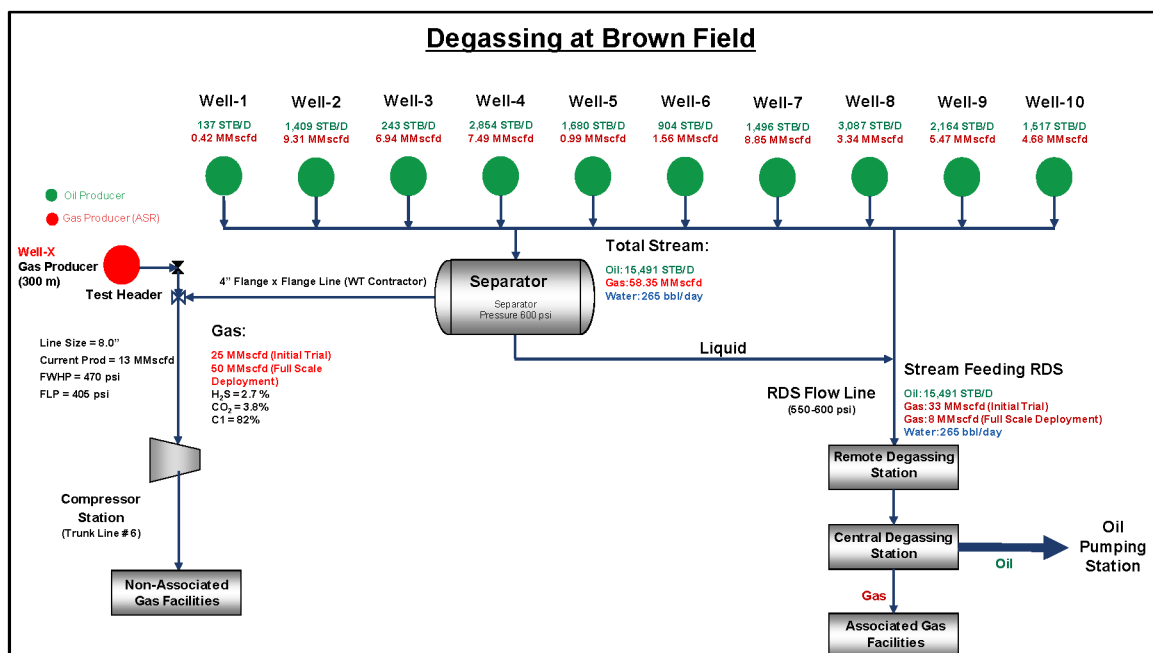


Figure 4—Overall Degassing Concept from Starting from Wells to Terminals

Table 1—Minipad-A Well Potential Data Based on Last Well Tests

Minipad-A Well Potential Data Based on Last Well Tests									
Well	Formation	Date	Choke	FWHP	Oil Rate	Gas	GOR	Water	Water Cut
			(%)	(psi)	(STB/D)	(MMscfd)	(scf/bbl)	(bbl/day)	(%)
Well-1	Reservoir-2	1-Dec-24	24	1,600	137	0.42	3,066	44	24
Well-2	Reservoir-1	1-Aug-24	52	2,705	1,409	9.31	6,608	0	0
Well-3	Reservoir-1	23-Mar-24	24	3117	243	6.94	28,547	0	0
Well-4	Reservoir-1	3-Aug-24	84	2,110	2854	7.49	2,624	0	0
Well-5	Reservoir-1	19-Dec-24	60	2,509	1,680	9.99	5,946	0	0
Well-6	Reservoir-1	5-Jul-24	32	2,131	904	1.86	2,058	61	6
Well-7	Reservoir-1	17-Dec-24	31	2,523	1,496	8.85	5,916	0	0
Well-8	Reservoir-1	12-Oct-24	76	1,645	3,087	3.34	1,082	69	2
Well-9	Reservoir-3	23-Nov-24	40	1,300	2164	5.47	2,528	35	2
Well-10	Reservoir-3	2-Dec-24	40	1,930	1517	4.68	3,085	56	4
Total					15,491	58.35		265	

Defining Scope of Work

The scope of work for the pilot degassing operation at Minipad-A was designed to ensure safe, efficient, and scalable implementation of portable gas-liquid separation. The key components of the SOW included:

Equipment Setup and Integration.

- Rig-up of a portable two-phase separator at the MPFM test header of Minipad-A, bypassing the MPFM to allow direct separation of gas and liquid streams.
- Connection of the liquid outlet to the existing 12" oil gathering line leading to RDS-B, and the gas outlet to the Well-X Gas Producer flowline using a 4" flange-to-flange piping system.

Well Routing and Testing Strategy.

- Minipad-A hosts 10 oil producers (7 Reservoir-1, 2 Reservoir-2, 1 Reservoir-3), all connected via a common header. Only one well at a time could be diverted through the test header for degassing.
- Phase-1 focused on Reservoir-1 wells with moderate to low H₂S content, excluding high-H₂S Reservoir-2 & 3 wells to ensure safe gas handling.
- Wells were added sequentially starting with the highest GOR producers, scaling up until the separator's handling capacity was reached (limited to 6,000 bbl/day of liquid and 25 MMSCFD of gas in Phase-1).

Gas Handling and Routing.

- The separated gas was routed to Well-X GP, which was connected to Compressor Station for delivery to Non-Associated Gas network.
- A mixed stream gas sample was analyzed confirming acceptable composition (H₂S = 2.7%, CO₂ = 3.8%, SG = 0.69, MW = 20.05) for routing to the depletion gas network.

Liquid Handling.

- The degassed liquid (oil and water) continued through the 12" line to RDS-B and onward to BCDS for final processing

Composition Modeling GAP Model for Degassing Optimization, Sensitivity Analysis and Strategic Insights

In complex production environments such as mature brown fields in the Middle East, accurate modeling of multiphase flow is essential for optimizing field performance and designing interventions like localized degassing. Traditionally, many network models have relied on the Black Oil Model, which simplifies fluid behavior by using fixed parameters for oil, gas, and water phases. While suitable for basic volumetric analysis, this approach lacks the granularity needed to simulate dynamic phase behavior, especially under varying pressure, temperature, and well combinations. In mature brown fields with complex production networks, accurate modeling of multiphase flow is essential for optimizing field performance and designing interventions such as localized degassing. One of the foundational elements in building a reliable network model using tools like GAP (Petroleum Experts) is the composition model, which defines the physical and chemical makeup of the produced fluids—oil, gas, and water—at the well level and across the network. The composition model is critical because it governs how fluids behave under varying pressure and temperature conditions throughout the network. It influences phase behavior, density, viscosity, and compressibility, which in turn affect flow dynamics, separator efficiency, and compressor performance. Without accurate compositional data, network simulations can yield misleading results, especially when modeling scenarios involving gas separation, reinjection, or routing to depletion gas networks.

In the context of the proposed localized degassing strategy, the composition model becomes even more vital. As gas is separated at pad or minipad level and routed to nearby depletion gas networks—typically connected to non-associated gas producers—the model must accurately reflect the gas composition to ensure compatibility with the receiving network. This includes matching molecular weight, CO₂ & H₂S contents, and other specifications required for reinjection or processing. Simultaneously, the liquid stream (oil and water) redirected back into the oil network must be modeled with updated composition to reflect the absence of separated gas. This affects downstream pressure profiles and flow assurance calculations. The composition model ensures that these changes are captured dynamically in the GAP simulation, allowing engineers to assess the impact of degassing on overall network performance.

To evaluate and optimize the proposed localized degassing strategy in a mature brown field in the Middle East, a detailed network model was constructed using GAP (Petroleum Experts). The model incorporated production data from nine wells, each contributing distinct oil and gas volumes. As illustrated in the schematic, these wells—ranging from Well-1 (137 STB/D, 0.42 MMSCFD) to Well-8 (2,618 STB/D, 7.41 MMSCFD)—collectively produce 15,491 STB/D of oil, 38.35 MMSCFD of gas, and 265 bbl/day of water.

Each well was modeled individually in GAP, capturing its unique flow rate and fluid composition. The wells were then grouped into different operational scenarios to simulate how routing selected wells through a portable separator—hooked up at the MPFM test header—would affect the overall network. The separator, operating at 500 psi, was modeled to split the stream into gas and liquid phases, with the separated gas diverted to a nearby depletion gas network connected to a non-associated gas producer, and the liquid redirected back into the oil network. The GAP model enabled extensive sensitivity analysis, allowing engineers to simulate various combinations of wells flowing through the separator at different choke sizes. This was essential because each well has a unique composition and flow behavior, and mixing them alters the final fluid characteristics. Although only one mixed gas sample was collected and used to calibrate the model, it provided a baseline for compositional tuning.

Through simulation, the model helped predict how the separated gas composition would interact with the non-associated gas network, ensuring compatibility in terms of pressure, calorific value, and processing

requirements. It also allowed evaluation of how the liquid stream, now stripped of gas, would behave when reintroduced into the oil network—impacting flowline pressure, pump sizing, and downstream separation efficiency. This modeling approach provided the flexibility to identify optimal well groupings for degassing, balancing gas removal efficiency with oil throughput. It also supported decision-making on which wells to route through the separator and which to leave on the conventional path, maximizing the value of the degassing solution while maintaining operational integrity.

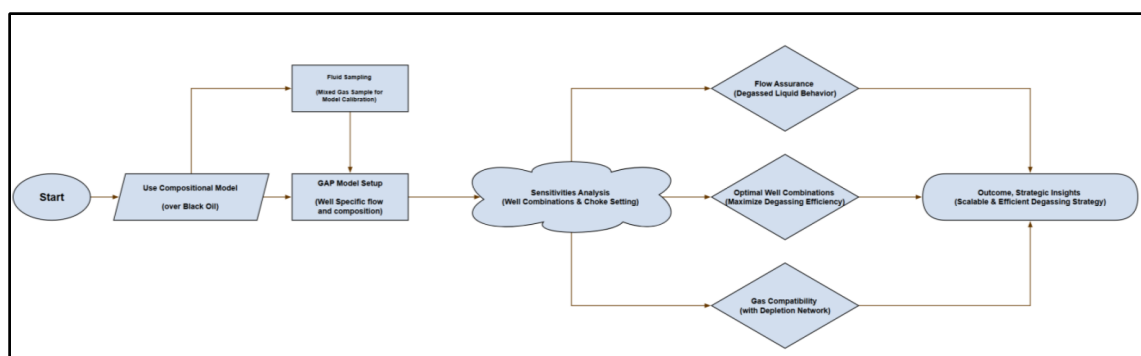


Figure 5—Workflow for Compositional GAP Network Model for Sensitivities Analysis and Strategy Insights

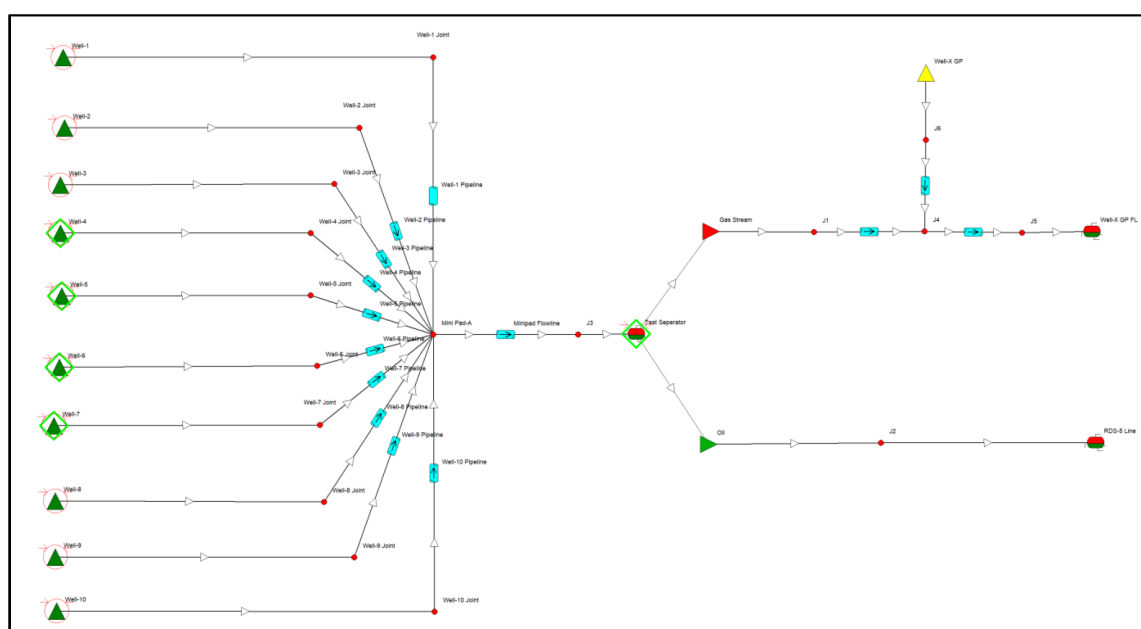


Figure 6—Workflow for Compositional GAP Network Model for Sensitivities Analysis and Strategy Insights

Operational Value and Added Benefits. *The operational value of compositional modeling lies in its ability to simulate how fluids behave under real-world conditions. It enables engineers to:*

- Predict phase behavior across the network with high fidelity.
- Assess separator performance based on actual gas-liquid ratios.
- Evaluate compatibility of separated gas with **non-associated gas networks** and **gas producers**, ensuring that reinjected or processed gas meets required specifications.
- Model the impact of gas removal on the liquid stream, including changes in pressure drop, pump sizing, and flow assurance.

Strategic Insights and Optimization. Through compositional modeling, we gained strategic insights into:

- How to balance gas removal efficiency with oil throughput.
- Decisions are required on which wells to route through the separator and which to leave on the conventional path. **Table-2** demonstrates how various sets of wells result in different fluid compositions, allowing for selection of well combinations compatible with downstream gas depletion network.
- How the degassed stream would behave under different operational scenarios.
- The impact of separator pressure on network stability.

Table 2—Effect on Fluid Composition of Separated Fluid by Routing Different Combination Wells

Fluid Composition at Separator (Well-2, 3, 4, 5 Online)			Fluid Composition at Separator (Well-1, 6, 9, 10 Online)		
Name	Liquid Comp. (%)	Accum. Gas (%)	Name	Liquid Comp. (%)	Accum. Gas (%)
N ₂	0.000308239	0.198613	N ₂	0.000148711	0.110651
CO ₂	0.0825262	4.6480087	CO ₂	0.0362817	2.1107771
H ₂ S	0.233893	3.5676351	H ₂ S	0.499434	7.4437416
C1	0.419338	73.343194	C1	0.338975	64.99534
C2	0.243092	6.5004957	C2	0.381515	10.724045
C3	0.647681	4.4164127	C3	0.861258	6.1175841
iC4	0.405661	1.0748513	iC4	0.488764	1.3647351
nC4	1.2828269	2.2957108	nC4	1.534048	2.8388111
iC5	1.3646234	0.901282	iC5	1.5779087	1.0861275
nC5	2.1890179	1.059573	nC5	2.4246613	1.2137009
C6	8.4420879	1.1756706	C6	7.3781955	1.0768181
C7-C12	17.043534	0.0315539	C7-C12	2.2696713	0.0165837
C13-C21	9.2670837	9.98E-06	C13-C23	1.058436	7.70E-06
C22-C80	4.6836442	4.72E-11	C24-C80	0.444799	4.90E-13
C7	4.413116	0.42312	C7-C11	31.374422	0.550731
C8	7.2644139	0.256258	C12-C19	17.896914	0.00102146
C9	6.1208338	0.0664915	C20-C80	9.2292866	8.56E-09
C10	8.4293038	0.030958	C7	1.8928089	0.188176
C11	6.2218934	0.0080118	C8	3.0528106	0.112835
C12	4.6075673	0.00170883	C9	2.5314783	0.0295468
C13	3.5706257	0.000361169	C10	3.4657688	0.0140175
C14	2.6835798	6.75E-05	C11	2.5529972	0.0037297
C15-C16	3.7593025	1.10E-05	C12	1.8892348	0.000808822
C17-C18	2.1483342	3.88E-07	C13	1.4637718	0.000172725
C19-C23	2.6823845	6.13E-09	C14	1.1000676	3.27E-05
C24-C72	1.7933269	0	C15-C16	1.5410146	5.45E-06
			C17-C18	0.880643	1.97E-07
			C19-C23	1.0995656	3.06E-09
			C24-C72	0.735121	0
Solution GOR	5063.2611	scf/STB	Solution GOR	2656.2914	scf/STB
CGR	197.50117	STB/MMscf	CGR	376.46472	STB/MMscf
Oil API	51.350248		Oil API	44.720198	
Gas Gravity	0.836945		Gas Gravity	0.894995	
Oil Molecular Weight	158.61676		Oil Molecular Weight	162.54793	
Accum Gas Molecular Weight	24.226794		Accum Gas Molecular Weight	25.90716	
Mole Fraction Oil	0.113926		Mole Fraction Oil	0.198815	
Mole Fraction Accum Gas	0.886074		Mole Fraction Accum Gas	0.801185	
Oil Molar Volume	0.584712	STB	Oil Molar Volume	0.577476	STB
Gas Molar Volume	378.88348	scf	Gas Molar Volume	378.88348	scf
Results of sending 1 mole of feed through separators			Results of sending 1 mole of feed through separators		
Volume of Oil	0.066614	STB	Volume of Oil	0.114811	STB
Volume of Gas	337.28405	scf	Volume of Gas	304.97118	scf
Weight of Oil	18.070611	lbm	Weight of Oil	32.316947	lbm
Weight of Gas	21.466727	lbm	Weight of Gas	20.756431	lbm

This approach established a data-driven framework for decision-making and facilitated the development of a degassing strategy consistent with upstream innovation objectives. The compositional model was used to predict gas composition, separator efficiency, and downstream impacts. It informed decisions on well selection and choke optimization, which led to diverting 25 MMSCFD of gas and increasing production by 7,000 BOPD from wells that were previously shut-in or choked back. The degassed liquid stream demonstrated improved flow assurance, and the separated gas complied with reinjection specifications. Relative to black oil modeling, the compositional approach offered operational adaptability, detailed modeling, and reliability.

Safety and Environmental Risk Assessments

The deployment of a localized degassing solution in a mature brown field in the Middle East was supported by a rigorous safety and environmental risk management framework. This included structured assessments such as HAZID, HAZOP, and Gas Dispersion and Radiation Studies, alongside continuous 24-hour operational supervision by a competent crew. These measures ensured that the solution was not only technically sound but also safe, reliable, and compliant with international operational standards.

HAZID: Early Hazard Identification. The HAZID (Hazard Identification) study was initiated during the conceptual design phase to systematically identify potential hazards associated with:

- Installing portable separators at pad and minipad levels.
- Routing separated gas to nearby depletion gas networks and non-associated gas producers.
- Reintegrating degassed liquids into the oil network.

This early-stage assessment helped flag risks such as overpressure scenarios, hydrocarbon leaks, venting hazards, and environmental exposure. Mitigation measures—including pressure control valves, emergency shutdown systems, and gas quality monitoring—were incorporated into the design to address these risks proactively.

HAZOP: Operability and Control Integrity. Following HAZID, a detailed HAZOP (Hazard and Operability) study was conducted to evaluate the operability of the degassing system under various operational scenarios. The HAZOP sessions involved multidisciplinary teams and focused on:

- Flow deviations due to fluctuating well rates and choke settings.
- Separator performance under mixed fluid compositions.
- Compressor loading and backpressure effects.
- Integration with MPFM headers and control systems.

The study revealed the need for dynamic control logic, monitoring of gas composition, and safeguards to prevent liquid carryover into the gas stream. These insights were used to refine the GAP model and validate compositional sensitivity simulations.

Gas Dispersion and Radiation Studies: Environmental and Personnel Safety. To assess environmental and human safety risks, Gas Dispersion and Radiation Studies were conducted. These studies modeled the behavior of separated gas in the event of a leak or venting scenario, considering wind patterns, ambient conditions, and proximity to personnel and equipment. Key outcomes included:

- Defining safe operating envelopes for separator and compressor installations.
- Establishing exclusion zones and emergency response protocols.

- Validating that radiation levels from flaring or venting remained within acceptable limits under both normal and upset conditions.

These studies ensured that the degassing solution met Company's stringent safety and environmental standards.

24-Hour Operational Supervision. To ensure safe and uninterrupted execution, the entire degassing operation was supervised 24 hours a day by a competent and trained crew. This continuous oversight was critical for:

- Monitoring real-time performance of separators and compressors.
- Managing well routing and choke adjustments.
- Responding promptly to any operational deviations or safety triggers.

The presence of a skilled crew ensured that the system operated within its design envelope and that all safety protocols were strictly followed.

The integration of HAZID, HAZOP, Gas Dispersion and Radiation Studies, and round-the-clock supervision reflects a holistic approach to safety and operational excellence. These measures provided the foundation for deploying a scalable, efficient, and environmentally responsible degassing strategy that aligns with upstream innovation and sustainability goals.

Field Execution Strategy

The degassing operation preparations began with site readiness checks, including layout validation, equipment mobilization, and stakeholder alignment. The portable separator was rigged up at the MPFM test header at Minipad-A, bypassing the flow meter to enable two-phase separation of gas and liquid. The liquid stream was routed to RDS via the existing oil network, while the separated gas was diverted to the Well-X Gas Producer flowline, which was connected to the depletion gas network. Safety protocols were strictly enforced, including flare pit construction based on dispersion and radiation modeling, installation of windsocks, and execution of fire and H₂S drills. Pressure testing of all connections was conducted prior to startup, and the green burner pilot system was kept lit throughout the operation. The degassing process began with the highest GOR well and gradually incorporated additional wells until the separator's capacity was reached. The operation was manned 24/7 by a competent crew, with continuous monitoring and data acquisition every 30 minutes to support real-time optimization and reporting.

This localized degassing strategy represents a practical and scalable solution to GOR-induced constraints in mature fields, aligning with regional upstream innovation initiatives and offering a replicable model for similar assets across the Middle East.

Benefits & Potential Impact

The degassing trial at Minipad-A yielded substantial production gains, enabling the recovery of up to 6,200 barrels of crude oil per day from previously shut-in wells that were constrained by high backpressure in the oil network. By implementing localized gas separation and routing the associated gas to a nearby depletion network, the operation unlocked significant flow potential and allowed for choke optimization across multiple wells. Economically, the impact is compelling: at a crude price of \$70 per barrel, the operation generated approximately \$13.02 million in monthly revenue and \$158.41 million annually. With a monthly rental cost of just \$150,000 for the portable separator and 24-hour crew, the net monthly profit stands at \$12.87 million, and the annual net profit reaches \$156.61 million. The financial indicators are compelling: the Return on Investment (ROI) stands at 87.01, and the monthly rate of return exceeds 85.80, demonstrating the high economic viability of the solution. The separator was sourced from existing well-

testing contracts, making it logistically simple and cost-effective to deploy. Encouraged by these results, the company has initiated a robust scale-up strategy aimed at unlocking \$2 billion in value from similar deployments across the field, with execution already underway and supported by operational learnings from this successful pilot.

Conclusions

The successful deployment of localized degassing at Minipad-A demonstrates a transformative approach to overcoming gas-handling constraints in mature brown fields. By integrating portable separators with existing MPFM infrastructure and routing separated gas to nearby depletion networks, the operation unlocked substantial oil production from previously shut-in wells while maintaining safety, compliance, and cost-efficiency. The compositional modeling and strategic execution framework enabled dynamic optimization and scalability, proving the viability of this solution across similar assets. With compelling economic returns and minimal infrastructure investment, this degassing strategy offers a replicable model for enhancing production reliability and operational flexibility in high-GOR environments—aligning with Company's upstream innovation goals and paving the way for broader field-level adoption.

Acknowledgments

The authors gratefully acknowledge the support and encouragement provided by the management team, whose strategic vision and commitment to innovation enabled the successful execution and publication of this work. Their endorsement of knowledge sharing and technical advancement has been instrumental in bringing this initiative to the broader professional community.

Nomenclatures

GOR	Gas Oil Ratio
MPFM	Multiphase Flow Meter
RDS	Remote Degassing Station
CDS	Central Degassing Station
NAG	Non-Associated Gas
BOPD	Barrel of Oil Produced per Day
WDW	Water Disposal Well
MOL	Main Oil Line
HAZID	Hazard Identification Study
HAZOP	Hazard and Operability Study
HSE	Health Safety and Environment
SOW	Scope of Work
MMSCFD	Million Standard Cubic Feet per Day
STB/D	Stock Tank Barrel per Day
FWHP	Flowing Wellhead Pressure

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